



Armada Kalamunda Group

Respiratory Virus Infection Prevention and Management Guideline

1. Applicability

Site	Operational Area	Applicable to
Armada Kalamunda Group	All areas	All workforce groups

2. Purpose

The purpose of this guideline is to provide details on the appropriate management of patients with respiratory viruses, and to minimise the risk of transmission to other patients, staff, and visitors. These guidelines apply to confirmed or suspected contagious respiratory viruses, including severe acute respiratory syndrome coronavirus (SARS-CoV, SARS-CoV 2), Middle East Respiratory Syndrome Coronavirus (MERS-CoV), influenza and other respiratory viruses that cause influenza-like illness (ILI), including but not limited to respiratory syncytial virus (RSV), human parainfluenza, adenovirus, rhinovirus and human metapneumovirus (HMPV).

3. Background

3.1 Symptoms

Respiratory viruses cause illnesses that affect the nose, throat and breathing passages including the lungs. These viruses' range in severity from mild cold or flu like symptoms to influenza and COVID-19. Respiratory virus infections have an acute onset with symptoms that include:

- Cough
- Breathing difficulty
- Sore throat
- Runny nose (rhinorrhoea) / nasal congestion

Additional symptoms can include:

- Fever ($\geq 37.5^{\circ}\text{C}$) or history of fever (e.g., night sweats, chills)
- Headache, myalgia, or fatigue
- Nausea, vomiting or diarrhoea.
- In the elderly consider, new or increased confusion, change in baseline behaviour, falling, exacerbation of underlying chronic illness.

3.2 Transmission

Transmission is via direct contact with an infected person, or indirect contact with hands, tissues or other items soiled by nasal or throat discharges. Some activities can increase the risk of

airborne transmission of disease such as aerosol generating procedures (AGPs) and aerosol generating behaviours (AGBs). Some viruses can also be spread through contamination of hands or objects with infected faeces (e.g., adenovirus).

4. Patient Management

4.1 Assessment and Screening

All patients attending AKG services should be assessed to determine if transmission-based precautions are required and whether respiratory virus testing should be considered. Early recognition and prompt isolation of patients who are suspected or known to have a respiratory virus infection is essential in reducing the risk of transmission.

Clinical judgement should be applied where there are alternative clinical explanations for symptoms.

4.2 Testing

In the hospital setting, RAT and/or PCR testing should be used to guide treatment (e.g., options for use of antiviral medication), and infection prevention and control management (e.g., use of transmission-based precautions, inform cohorting options).

- COVID-19 is listed as a notifiable disease under the Public Health Act 2016 and hospital clinicians must notify the Communicable Disease Control Directorate (CDCD) of all positive cases.
- The WA Department of Health no longer requires registration of positive RAT results.

Refer to [Specimen Collection and Pathology Results Procedure](#)

4.3 Accommodation

- A single room with dedicated bathroom/ensuite should be allocated as per the [Standard and Transmission-based Precautions Procedure](#)
- The Droplet precautions “**Green Card**” should be prominently displayed on the door of isolation rooms.
- Patients may be cohorted with the same laboratory confirmed type and strain of respiratory virus. Only confirmed respiratory virus cases without other infectious conditions (e.g. gastroenteritis) or Micro-Alerts can be cohorted.
- Children <12 months should not be cohorted.
- Demand for isolation rooms can often exceed the number of rooms available within the hospital. Refer to [Single Room Prioritisation and Cohorting of Inpatients on Transmission-based Precautions Procedure](#)

4.4 Personal Protective Equipment (PPE)

It is acknowledged that many respiratory virus illnesses exist on a continuum between droplet and airborne transmission. HCWs providing care to those patients who are confirmed or suspected of respiratory virus infection must wear the following PPE at a minimum:

- A fit tested particulate filter respirator (PFR) and protective eyewear (face shield or goggles)

Standard precautions always apply, and healthcare workers should continue to

- Wear gloves if there is risk of exposure to blood and/or bodily fluids

- Wear an apron/long-sleeved gown to protect their uniforms during:
 - Significant close physical contact with the patient
 - When providing care during aerosol generating procedures/behaviours

Table 1: Healthcare worker personal protective equipment

Donning PPE		Removing PPE	
	Perform Hand hygiene Remove jewellery, ensure bare below the elbows, Perform hand hygiene		Removing and disposing of gloves - Remove by gripping at cuffs and slowly peeling one glove off at a time –Immediately dispose of gloves in appropriate waste bin
	Fitting gown - Put the gown on so your front and arms are fully covered, and gown opening is at the back		Perform Hand hygiene
	N95 or P2 particulate filter respirator (PFR) Perform fit check		Removing and disposing of gown - With clean hands untie gown. Touching only the inside of the gown, pull from the neck and shoulders, then arms, turning inside out as gown is removed. Roll gown into bundle and discard in appropriate waste.
	Fitting goggles or face shield – Position goggles over eyes or face shield over face, adjust if necessary		Perform Hand hygiene
	Perform Hand hygiene Fit gloves over cuff of gown – Fit gloves, adjusting at the cuffs.		Removing and disposing of goggles/face shield - Remove eyewear by handling sides or back only. Discard or reprocess with disinfectant wipes, store dry
			Perform Hand hygiene
			Removing and disposing of mask - Remove mask by only handling at ties. Discard in appropriate waste.
			Perform Hand hygiene

5. Duration of isolation precautions

Note: Day of symptom onset or positive test result is Day = 0

Virus/Illness	Immunocompetent patient	Significantly immunocompromised patient
Adenovirus (causing respiratory symptoms)	Precautions continue for the duration of symptoms	
Influenza	Precautions continue for 5 days duration following onset of illness OR 72 hours after commencing Oseltamivir antiviral medication	Discuss with Clinical Microbiologist / ID Physician before removing precautions
Middle East Respiratory Syndrome (MERS-CoV)	Continue precautions until 24 hours after the resolution of symptoms	
SARS-CoV-2 (COVID-19)	Precautions continue for 7 days duration following onset of illness or until resolution of symptoms, whichever is longer	Precautions continue for 10 days duration following onset of illness and 2 negative RAT taken on separate days
Coronavirus strains (not COVID-19) Human metapneumovirus (HMPV) Parainfluenza Pneumonia (viral) Respiratory syncytial virus (RSV) Rhinovirus Influenza-like illness, untested or inconclusive result	Precautions continue for 5 days duration following onset of illness, OR resolution of symptoms, whichever occurs first	Discuss with Clinical Microbiologist / ID Physician before removing precautions

6. Other precautions

6.1 Patient Transport

Patient transfers within the hospital can be conducted as per standard procedures with the addition of:

- Encouraging the patient to wear a surgical mask for the duration of transfer

- If on oxygen therapy, the patient should be transitioned to nasal prongs with a surgical mask over the top. If the patient is unable to transition to nasal prongs, a surgical mask should be placed over the Hudson mask prior to transport
- The receiving department must be notified of pending arrival
- HCW involved in the transfer should wear a PFR & eye protection at a minimum
- The HCWs accompanying the patient must don fresh PPE prior to transfer

6.2 Management of the deceased

- HCWs should wear PFR and eye protection in addition to standard precautions when preparing the body for transport.
- Family members may view the deceased. After viewing, bereaved should immediately wash their hands or use an alcohol-based hand rub.
- For transportation, secure the body in a leak-proof body bag to prevent leakage of body fluids.
- Inform mortuary staff of the deceased personal infection status prior to transfer.
- Mortuary HCWs are to follow routine guidelines for management of the deceased.

6.3 Visitors

All visitors are strongly encouraged to wear a surgical mask when visiting high-risk areas or visiting vulnerable/immunocompromised patients. Surgical masks should be available for all visitors to access at hospital entry points and ward reception areas.

6.4 Air purifier units

The use of air purifier units (APUs) can assist with the reduction of circulating infectious particles and therefore likelihood of transmission of respiratory infections.

Placement of APUs may include the following strategies:

- Single rooms that are not NPIR
- Multi-bed rooms used to cohort COVID-19 or ILI patients, including areas such as Dialysis Units
- Corridors and staff areas of high-risk areas
- During AGPs/AGBs
- Areas of high flow traffic e.g. Emergency Department and waiting rooms
- Other applications as advised by IP&M

Refer to [Air Purifiers \(HEPA filters\) Management Procedure](#) for further recommendations and process for requesting a unit.

6.5 Other standard and transmission-based precautions requirements

- Refer to [AKG Patient Support Services Cleaners Manual](#) for guidance on daily cleaning and disinfection of isolation room.
- Linen management as per standard precautions, outlined in the [Linen Management Policy](#)
- Normal waste management requirements apply. PPE can be disposed of in general waste unless heavily contaminated/soiled by bodily fluids.

7. Staff management

It is strongly recommended all HCWs are vaccinated against respiratory viruses such as COVID-19 and seasonal Influenza.

Staff with symptoms of acute respiratory virus infection, regardless of whether they have recently recovered from previous respiratory virus infection, including COVID-19, should undertake a SARS-CoV-2 RAT or seek PCR testing and self-exclude from work as per [EMHS Staff Furloughing Guide](#).

Staff who are **asymptomatic COVID-19 Close Contacts** should wear a surgical mask at a minimum, while at work, for 7 days from when they first meet the definition of a [close contact](#), daily RAT, and self-monitor for symptoms of acute respiratory virus infection.

Staff who are **symptomatic COVID-19 Close Contacts** are discussed in the [EMHS Staff Furloughing Guide](#).

8. Legislative context

The following documents are required to give effect to this policy:

- *Public Health Act 2016*
- *Public Health Regulations 2017*
- Health Department of Western Australia (HDWA) [Health Care Worker Immunisation Policy OD 0388-12](#)
- HDWA [Healthcare Associated Infection Surveillance in Western Australia Policy MP 0108-19](#)
- East Metropolitan Health Service (EMHS) [Infection Prevention and Management EMHS: 34](#)

9. Supporting information

The following documents support the implementation of this policy:

- [Hospital Outbreak Management Policy AKG-POL-0334-16](#)
- [Standard and Transmission-based Precautions Procedure AKG-PRO-0513-24](#)
- [Single Room Prioritisation and Cohorting of Inpatients on Transmission-based Precautions Procedure AKG-PRO-0094-24](#)
- [Hand Hygiene Policy AKG-POL-0187-23](#)

10. Compliance, monitoring and evaluation

Compliance against this policy will be evaluated with results from clinical audits, IPM investigations, Root Cause Analysis and reporting of incidents into the Datix Clinical Incident Management

11. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 3 – Action 3.5 Standard and transmission-based precautions
- Standard 3 – Action 3.7
- Standard 3 – Action 3.8 Hand hygiene

12. References

1. Government of Western Australia (WA), Department of Health (2023) [COVID-19 Infection Prevention and Control in Western Australian Healthcare Facilities](#)
2. Government of WA, Department of Health (2023) [Respiratory Protection Program for Western Australian Healthcare Facilities](#)
3. Government of WA, Department of Health (2023) [Personal Protective Equipment in Healthcare Facilities Policy](#)
4. Australian Commission on Safety and Quality in Health Care (2019) [Australian Guidelines for the Prevention and Control of Infection in Healthcare](#)
5. RPBG (2023), Respiratory Virus Infection Prevention & Management Clinical Guideline

13. Glossary

The following definitions are relevant to this policy. Term	Definition
Aerosol generating behaviour (AGB)	Behaviours that are likely to generate higher volumes of respiratory secretions and thus increase the risk of transmission via aerosols e.g., shouting, spitting, screaming, women in active labour who exhibit heavy breathing and panting.
Aerosol generating procedure (AGP)	Procedures that promote the generation of fine airborne particles (aerosols) that may result in the risk of airborne transmission of disease including but not limited to: • Insertion or removal of endotracheal tube • Intentional or inadvertent disconnection/reconnection of closed ventilator circuit • High frequency oscillatory ventilation (HFOV) • Open oropharyngeal or tracheal suctioning • Upper respiratory instrumentation or surgery e.g., bronchoscopy, tracheostomy, ear nose throat surgery • Surgical or post-mortem procedures on respiratory tract involving high-speed devices. • Intercostal catheter insertion for relief of pneumothorax • Thoracic surgery that involved entering the lungs • Dental/oral procedures utilising equipment that generates aerosols e.g., ultrasonic scalers and high-speed handpieces. Other procedures can generate respiratory aerosols include: • Nebulisers • Manual or non-invasive ventilation ○ Bi-level positive airway pressure ventilation (BiPAP) ○ Continuous positive airway pressure ventilation (CPAP) • Collection of induced sputum • High flow nasal oxygen (HFNO) • Diagnostic instrumentation of the upper digestive tract, including transoesophageal echocardiography (TOE) • Cardiopulmonary resuscitation (CPR)
Air purifier unit (APU)	A device which removes contaminants from the air in a room to improve indoor air quality by filtering through a prefilter and / or a HEPA filter with or without a prefilter.
COVID-19 (Coronavirus disease 2019)	A contagious respiratory illness caused by the SARS-CoV-2 virus

Negative Pressure Isolation Room (NPIR)	A room in which the air pressure differential between the room and the adjacent indoor airspace directs the air flowing into the room i.e. room air is prevented from leaking out of the room and into adjacent areas such as the corridor
Particulate Filter Respirators (PFR)	Respirators that filter at least 94 percent of 0.3-micron particles from the air. PFRs are used when implementing airborne precautions. Both P2 and N95 respirators are appropriate for use with airborne precautions
Polymerase chain reaction (PCR) test	A laboratory DNA amplification process used for detection of viruses, bacteria, and diagnosis of genetic disorders.
Rapid antigen test (RAT)	A point of care test that directly detects the presence of a pathogen-associated antigen in a specimen. They are commonly used as a screening test for COVID-19.
Significantly immunocompromised	Immunocompromising conditions include: • Primary or acquired immunodeficiency ○ Haematologic neoplasms: leukaemia's, lymphomas, myelodysplastic syndromes ○ Post-transplant: solid organ (on immunosuppressive therapy), haematopoietic stem cell transplant (within 24 months) ○ Immunocompromised due to primary or acquired (AIDS) immunodeficiency ○ HIV with CD4 T-lymphocyte count <250μL or those with a higher count unable to be established on effective therapy. ○ Other significantly immunocompromising conditions • Immunosuppressive therapy (current or recent) ○ Chemotherapy, whole body radiotherapy or total lymphoid irradiation ○ High-dose corticosteroids ($\geq$20mg of prednisone per day, or equivalent) for $\geq$14 days ○ Selected other potent immunosuppressive therapies

14. Document owner

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This guideline will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
V4	21/10/2024	21/10/2027	3. Background – Focus on relevant on symptoms, modes of transmission etc 4. Pt management – included policy links and testing requirements 5. Duration of precautions – not previously included 6. Other precautions – expanded to provide more details 7. Staff management – removed HIN1 table is no longer relevant. COVID furlough guideline linked

15. Authorisation

This guideline has been authorised and issued by the Director Nursing and Midwifery.

Authorised by	Helen Fullarton – Director Nursing and Midwifery
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